

AssetFlow

Texnik Strategiya

Higgsfield va Motion Factory pluginlarining tizim-bo'yicha 100% texnik tahlili. Qaysi tizim nega kuchli, qaysi texnologiyada qurilgan, va AssetFlow'ga nima qo'shish kerak — ipidan ignasigacha.

Tech-stack tahlili

14 ta tizim

Arxitektura metodologiyasi

Nima qo'shish + nega

Maqsad arxitektura

Tayyorlandi: 2026-yil 13-iyun · Loyiha: creative-tools-saas (AssetFlow)

Manbalar: Higgsfield v1.0.14 (Bolt CEP) · Motion Factory v4.9.5 (66 Web Component) · AssetFlow v1.1.1 (monorepo)

Har qanday qisqacha kod, endpoint va fayl havolalari haqiqiy manba tahlilidan olingan.

1. Boshqaruv xulosasi (texnik)

Savolingiz: "Texnik jihatdan ikkala raqibdan nimani olishim kerak, qaysi tizimlari menikidan kuchli va nega, qaysi bo'limlarni qo'shish kerak — to'liq strategiya." Quyida har bir tizim alohida tahlil qilingan.

ENG MUHIM STRATEGIK XULOSA — bitta harakat 3 muammoni hal qiladi

AssetFlow pluginini

Bolt CEP

(Higgsfield ishlatadigan framework) ga ko'chirish bir vaqtning o'zida:

(1)

ZXP paketlash+imzolashni beradi (vite-cep-plugin ichida ZXP build bor — bu sizning №1 blokeringiz),

(2)

typed evalTS ko'prigini beradi (injection xavfini tugatadi),

(3)

zamonaviy komponentli UI beradi. Uchta eng katta texnik qarzni bitta migratsiya yopadi.

Umumiy hukm

Soha	Kim kuchli	Sabab (qisqa)
Backend & ma'lumotlar bazasi	AssetFlow	Express+Prisma+PostgreSQL — MF'da WordPress, HF'da yopiq SaaS
Auth xavfsizligi	AssetFlow	PluginToken (32-bayt, DB, server-validatsiya) » MF date-token; HF darajasiga yaqin
Contributor/moderatsiya	AssetFlow	Ikkala raqibda umuman yo'q
Plan gating (server-side)	AssetFlow	localStorage'dan chetlab bo'lmaydi; MF gating'ni localStorage'da saqlaydi
CEP↔ExtendScript ko'prigi	Higgsfield	typed evalTS vs sizning xom string evalScript (injection)
Xato monitoringi	Higgsfield	Sentry vs sizda hech narsa yo'q (faqat console)
Qidiruv	Motion Factory	Algolia (typo-tolerant, tez) vs sizning findMany +client filter
Auto-update	Motion Factory	To'liq avtomatik (download+extract+self-replace) vs sizda yo'q
Build/paketlash	Ikkalasi	Bolt CEP / npm — sizda plugin build yo'q, ZXP yo'q
AI funksiyalar	Ikkalasi	HF: 36 model; MF: ovoz+SFX — sizda yo'q
Xavfsizlik (umumiy)	AssetFlow	MF: 7 jiddiy zaiflik; siz allaqachon to'g'ri yo'ldasiz

Bir jumlada: Sizning poydevoringiz (backend, auth, biznes-mantiq) ikkala raqibdan kuchli. O'rtda qolgan joy — plugin qatlamining yetukligi: build/paketlash, ko'prik xavfsizligi, monitoring, qidiruv, auto-update va AI. Bularning hech biri qayta yozishni talab qilmaydi — faqat ustiga qurish.

2. Tech-stack va arxitektura metodologiyasi

Har bir plugin tubdan boshqa metodologiyada qurilgan. Buni tushunish — nima ko'chirib bo'lishini aniqlaydi.

Higgsfield Bolt CEP · Vite + React 19 + TS + Tailwind v4 Metodologiya: Yagona kompilyatsiya qilingan SPA. Vite bitta <code>main-*.cjs</code> (71k qator, minified) + <code>init-cep-*.cjs</code> ga to'playdi, source-map bilan. ExtendScript: bitta <code>jsx/index.js</code> (2750 qator), <code>papro</code> + <code>aeft</code> namespace. Modul: <code>index.html</code> 'da maxsus AMD/require shim. State: TanStack React Query (server-state), <code>evalTS</code> bridge.	Motion Factory Vanilla JS · 66 ta pixflow-* Web Component npm paket Metodologiya: Mikro-modulli. Har funksiya alohida npm paket → <code>customElements.define()</code> bilan Web Component. Kompozitsiya: <code>main.js</code> paketlarni <code>require()</code> qiladi; <code><mf-templates></code>, <code><mf-voiceover></code> kabi teglar. Aloqa (3 kanal): CustomEvent + umumiy <code>localStorage["init-data"]</code> + window event-bus (<code>enable-toolbar</code> ...). Build: faqat SASS kompilyatsiya; JS xom.	AssetFlow (sizniki) Monorepo · Express 5 + Prisma + Next + statik Metodologiya: npm workspaces. <code>apps/api</code> (Express+TS), <code>apps/web</code> (Next 15), <code>packages/database</code> (Prisma 6.8), <code>packages/assetflow-studio</code> (statik HTML). Plugin: <code>plugins/after-effects-cep</code> — build yo'q, xom HTML/JS/JXS. <code>host.jsx</code> 2229 qator, ~110 funksiya. Deploy: API→Render, Studio→Vercel, lokal→PM2.
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Ko'chirilishi bo'yicha xulosa

Higgsfield'dan

framework darajasidagi narsalarni olish mumkin (Bolt CEP, typed bridge, Query-cache, Sentry).

Motion Factory'dan

alohida tizimlarni olish mumkin (Algolia, auto-update, AI-proxy formulasi) — lekin

uning xavfsizlik anti-naqshlarini emas

. AssetFlow'ning monorepo backend'i ikkalasidan ustun — uni saqlang.

3. Tizim-bo'yicha chuqur taqqoslash

14 ta tizim. Har biri: HF qanday qiladi · MF qanday qiladi · siz qanday qilasiz · **hukm + tavsiya**.

Ⓢ Build & paketlash

Raqiblar kuchli

Higgsfield

Bolt CEP = Vite. ZXP imzolangan paket. Source-map shipped. Hot-reload dev.

Motion Factory

npm + SASS build. ZXP imzolangan. 66 paket versiyalanadi.

AssetFlow

Build yo'q. Xom fayllar. **ZXP yo'q** — faqat `install-cep.sh` (imzosiz, dev-only).

Tavsiya: **N#1 blocker**. Bolt CEP'ga ko'ching (`vite-cep-plugin` ZXP imzolashni o'z ichiga oladi) **YOKI** minimal yo'l: `ZXPSignCmd` + self-signed `.p12` + `scripts/build-zxp.sh`. Bunsiz oxirgi foydalanuvchi o'rnatolmaydi.

Ⓢ CEP ↔ ExtendScript ko'prigi

Higgsfield kuchli

Higgsfield

Typed
`evalTS/listenTS/dispatchTS`.
Argumentlar `JSON.stringify` → `try/catch`
`ExtendScript` → `Promise`. Namespace
`host["ai.higgsfield.cep"].fn()`.

Motion Factory

Xom `evalScript`, qo'lda string
`concat`. **Fayl yo'llari to'g'ridan-to'g'ri interpolatsiya** → injection.

AssetFlow

Xom string:
`evalScript('importSingleSceneFromAep('+JSON.stringify(cfg)+')')`.
Typed wrapper yo'q. `host.jsx` `evalScript` limitidan oshib `$.evalFile` fallback ishlatadi.

Nega muhim: Xom string ko'priki = Ⓢ injection xavfi (zararli nom/yo'l `ExtendScript` bajaradi), Ⓢ type xavfsizligi yo'q, Ⓢ har chaqiruvni qo'lda yozish — xatoga moyil. Higgsfield naqshi sanoat standarti.

Tavsiya: `evalJSX(fn, ...args)` typed wrapper qo'shing (Bolt CEP'ga ko'chsangiz tekin keladi). Argumentlarni faqat JSON sifatida uzating, `ExtendScript` ichida `JSON.parse` qiling — string `concat` qilmang.

Ⓢ Auth & sessiya

AssetFlow kuchli (1 kamchilik bilan)

Higgsfield

OAuth 2.0 + PKCE (S256) + Clerk. `refresh_token`
bor. Token `~/higgsfield-auth-*.json` (fs).
`Bearer id_token`.

Motion Factory

date-token ddmmyy (taxmin qilinadi). `WebSocket`
`login`. `localStorage`'da token+email. Hardcoded
`pxAPIKey`.

AssetFlow

JWT (HS256, 7d) + `PluginToken` (32-bayt random, 30d, DB, server-validatsiya). `prefs.json` (disk).
Refresh yo'q.

Olish (HF'dan): `refresh_token` flow — 30 kun tugaganda majburiy qayta-login o'rniga jim yangilash. **Tuzatish:** `apply-ae-prefs` hardcoded fayl yo'liga bog'liqligi (server+plugin bir mashinada deb faraz qiladi — faqat dev'da ishlaydi). **OLMANG (MF'dan):** `date-token`, `localStorage` auth, hardcoded key.

Ⓢ API dizayni & klient

Aralash

Higgsfield

`fetch` + TanStack Query (cache, polling
`refetchInterval`, `retry-lanes`, `abort-signal`).
Base `fnf.higgsfield.ai`.

Motion Factory

Xom `XMLHttpRequest`, WordPress REST (23+ endpoint), `wp-json/pixflow/*`.

AssetFlow

Express 5, RESTful, Zod validatsiya, Prisma xato-mapping (P2025→404, P2002→409). Klient: xom `fetch`, har filtr/qidiruvda `innerHTML` qayta-chizish.

Server: sizniki kuchli (Zod + tipli xato-handling). Klient'da olish (HF'dan): cache + polling abstraktsiyasi (job-status uchun) va `innerHTML` teardown o'rniga incremental DOM yangilash. Vanilla'da ham kichik query-cache yozish mumkin.

Ⓢ Qidiruv

Motion Factory kuchli

Higgsfield

Server-side (galereya AI generatsiyaga qaratilgan, katalog qidiruv kam).

Motion Factory

Algolia (appld `WS11ARD05G`). Indekslar
`pixflow_elements` + `_trending`. Typo-tolerant, `optionalFilters`, kategoriya-query AND-birikma.
Klient-side, bir zumda.

AssetFlow

findMany + klient-side `nav/cat` filter.
Relevance scoring qo'lda. Typo-tolerance yo'q, scale bo'lmaydi.

Nega muhim: Katalog o'sgani sayin `findMany` +JS-filter sekinlashadi, relevance/typo-tolerance bermaydi. MF 10k+ shablonni bir zumda qidiradi.

Tavsiya (bosqichma-bosqich): arzon → PostgreSQL `tsvector` / `pg_trgm` (sizda allaqachon Postgres bor). Keyin → Meilisearch/Typesense (self-host, arzon) yoki Algolia (bepul 10k so'rov/oy). **Diqqat:** MF'dek search-key'ni klientga hardcode qilmang — faqat search-only key, indeks faqat published yozuvlar.

📄 Yuklab olish (download) quvuri

AssetFlow kuchli

Higgsfield

`fetch` → `fs.write` → `cache.os.tmpdir`.

Motion Factory

POST signed-url → `https.get` stream → progress (content-length) → `AdmZip.extractAllTo` → library API.

AssetFlow

`https.get` qo'lda redirect (8 limit) → **cross-origin'da Auth header'ni olib tashlaydi** (token leak'dan saqlaydi) → shell `unzip -o` → `.aep/.mogrt` ajratish → selective scene download.

Sizniki eng xavfsiz — redirect'da Auth-header strip'i MF'da yo'q. **Kichik yaxshilash:** shell `unzip` o'rniga `adm-zip` (MF kabi) — system `unzip` mavjudligiga bog'liqlikni yo'qotadi (Windows'da yo'q).

📄 Adobe'ga import

Aralash

Higgsfield

`evalTS("importMediaFromPath", ...)`. Track Matte / nested sequence avtomatlashtirish.

Motion Factory

5 app: PPRO `importMGT / importFiles`, AEFT `ImportOptions`, LUT `ADBE Apply Color LUT2`, PHXS/ILST `open(File)`.

AssetFlow

`host.jsx` 110 funksiya — eng katta native sirt. Async open-project navbati (`app.open()`) bloklanishini chetlab o'tadi, undo-group, item-ID diff bilan yangi comp topish.

Sizning `host.jsx` chuqurligi raqiblardan ustun (async navbat — aqlli yechim). **Olish (MF'dan):** ko'p-app host-detection naqshi (`getHostEnvironment().appName` switch) — kelajakda PPRO/PS qo'shsangiz. **Tuzatish:** evalScript limitini typed bridge + `$.evalFile` bilan to'g'ri boshqaring.

📄 Fayl saqlash (storage)

AssetFlow kuchli

Higgsfield

S3 pre-signed URL, 2-bosqichli upload (confirm).

Motion Factory

CDN to'g'ridan-to'g'ri (link share qilinishi mumkin).

AssetFlow

Cloudflare R2 (AWS SDK v3). `lib-storage` stream-upload (partSize 8MB, peak ~32MB — 3GB faylda ham). Signed URL TTL 1h. Pack API orqali gated → 302 R2.

Sizniki production-darajasi (OOM-fix, stream-upload). **Bitta tuzatish:** thumb/preview public R2 URL (URL bilan hammaga ochiq). Agar muhim bo'lsa — preview'ni ham signed qiling yoki ochiqqligini ataylab qabul qiling.

📄 Subscription / billing / kreditlar

AssetFlow kuchli (1 bug bilan)

Higgsfield

Kredit tizimi + `/balance` API. Generatsiya kredit yechadi.

Motion Factory

localStorage'da `subscription_data` (**klientda — chetlab bo'ladi**). `ai_token_count/limit`.

AssetFlow

FREE (15 dl/oy + 10 import umrbod) / PRO. **Server-side gate** (`checkDownloadAllowed`) — localStorage'dan chetlab bo'lmaydi. Stripe webhook (raw body).

Sizning server-side gating'ingiz MF'dan tubdan xavfsiz. KRITIK BUG: `PLUGIN_ALLOW_PRO_WITHOUT_STRIPE="true"` production'da o'rnatilgan → hozir har kim `PATCH /plan` bilan bepul PRO oladi. Buni `false` qiling. **Quirk:** download oylik, import umrbod — assimetrik, hujjatlashtirib qo'ying.

📄 Analytics & xato monitoringi

Higgsfield kuchli

Higgsfield

Sentry (66 ref, captureException, breadcrumb, **sensitive-data filter regex**), Amplitude (309 ref, batch 10s), GrowthBook.

Motion Factory

GA4 (`px-client`). **Discord webhook** orqali xato hisoboti (leaked credential).

AssetFlow

Sentry yo'q. Amplitude yo'q. Faqat `console.*` + 500-yozuvli file-log + DB-aggregate analytics.

Nega muhim: Hozir production'da nima sinishini bilmaysiz — foydalanuvchi panel qotib qolsa, sizga signal kelmaydi. HF har xatoni stack-trace + breadcrumb bilan ko'radi.

Olish (HF'dan): 🕒 Sentry (bepul 5k event/oy) — panel + API. 🕒 Sentry'dan oldin sensitive-data filter regex (`/ (authorization|token|secret|password|api[_]?key)/i`) — token leak'dan saqlaydi. 🕒 Product analytics — Amplitude yoki PostHog (self-host).

11 Feature flags

Higgsfield kuchli

Higgsfield

GrowthBook (39 ref) — runtime'da feature toggle, A/B test, sekin rollout.

Motion Factory

Yo'q (init-data'dagi flag'lar).

AssetFlow

Yo'q. Faqat xom env var (`PLUGIN_ALLOW_PRO_WITHOUT_STRIPE`) — deploy talab qiladi.

Olish (HF'dan): GrowthBook (bepul, self-host mumkin) yoki PostHog flags. **Nega:** AI funksiyalarni xavfsiz, foydalanuvchi-guruhiga rollout qilish; env-var o'rniga deploy'siz toggle.

12 Auto-update

Motion Factory kuchli

Higgsfield

`manifest.json` polling → versiya solishtirish → brauzerda download link ochish.

Motion Factory

To'liq avtomatik: GET `get-plugin-latest-version-tag` → download zip → `rmdir` + `AdmZip.extractAllTo` → o'zini almashtirish.

AssetFlow

Yo'q. Manifest versiyasi `1.1.1` statik. Yangilash = `install-cep.sh` qayta ishga tushirish.

Nega muhim: Hozir tuzatish yoki yangi funksiyani foydalanuvchiga yetkaza olmaysiz — ular o'rnatilgan versiyada qotib qoladi.

Olish (MF'dan): versiya-tekshiruv endpoint (GET `/api/plugin/version`) + panel ochilishida solishtirish + notification (yoki MF'dek to'liq avtomatik download+extract). **Tuzatish (MF bug'i):** `res.tag_name > current` string solishtiruvi noto'g'ri (`"1.10" < "1.9"`) — **semver** ishlatish.

13 State boshqaruvi / realtime

Aralash

Higgsfield

TanStack Query cache + `invalidateQueries`. State sync `dispatchTS` event bilan. fs'da view-state.

Motion Factory

3 kanal: CustomEvent + localStorage init-data + window-bus.

AssetFlow

SSE upload-progress (**in-memory Map**, 25s ping). Rate-limit, SSE, file-log — **hammasi single-process**.

Sizning SSE'ingiz raqiblardan yaxshi (ularda realtime yo'q). **Diqqat:** in-memory state Render restart'da yo'qoladi va horizontal-scale'da sinadi. **Tavsiya:** ko'p instansiyaga o'tganda Redis (rate-limit + SSE pub/sub + cache).

14 Xavfsizlik holati

AssetFlow kuchli

Higgsfield

PKCE, sensitive-data filter, faqat `--enable-nodejs` `--mixed-context`. Yaxshi.

Motion Factory

7 jiddiy zaiflik: `--disable-web-security`, hardcoded pxAPIKey, date-token, `exec(curl)` injection, localStorage auth, Discord webhook, evalScript path-injection.

AssetFlow

Server-side gate, redirect'da Auth-strip, JWT_SECRET prod-tekshiruv. `--disable-web-security` faqat Admin panelda.

Siz MF'dan tubdan xavfsiz. Olib tashlash kerak: Admin paneldagi `--disable-web-security` (kerak emas bo'lsa). **Olmang (MF):** barcha 7 anti-naqsh. **Tuzatish:** evalScript injection (typed bridge bilan), Stripe-bypass flag, preview public-URL.

4. Qaysi tizimlar mendan kuchli — va NEGA

To'g'ridan-to'g'ri savolingizga javob. Har biri texnik sabab bilan.

Higgsfield'da kuchli (4 ta)

Tizim	Nega kuchli (texnik)	Sizga ta'siri
Typed bridge (evalTS)	JSON.stringify serializatsiya + try/catch + Promise. Injection yo'q, type-safe, qayta ishlatiluvchi.	Sizning xom string evalScript — injection xavfi + qo'lda yozish
Sentry monitoring	Har xato stack-trace + breadcrumb + sensitive-data filter bilan markazda.	Sizda 0 ko'rinish — production'da nima sinishini bilmaysiz
TanStack Query	Avtomatik cache, polling, retry-lane, abort-signal, invalidation.	Sizda innerHTML teardown har keystroke'da — jank
GrowthBook flags	Runtime feature-toggle, A/B, sekin rollout — deploy'siz.	Sizda faqat env-var — har o'zgarish deploy talab qiladi

Motion Factory'da kuchli (2 ta)

Tizim	Nega kuchli (texnik)	Sizga ta'siri
Algolia qidiruv	Typo-tolerant, optionalFilters, bir zumda klient-side, 10k+ yozuvda tez.	Sizning findMany+JS-filter scale bo'lmaydi, typo-tolerance yo'q
Avtomatik auto-update	Versiya-tekshiruv → zip download → o'zini almashtirish. Foydalanuvchi hech narsa qilmaydi.	Sizda yo'q — tuzatishni yetkaza olmaysiz

Ikkalasida ham bor, sizda yo'q (2 ta)

Tizim	Nega muhim	Manba
Build + ZXP paketlash	Imzolangan ZXP'siz oxirgi foydalanuvchi o'rnatma olmaydi — bu sizning №1 blokeringiz.	HF (Bolt CEP), MF (npm)
AI funksiyalar	"creative-tools" pozitsiyasi uchun zarur; yangi daromad (kreditlar); raqobat pariteti.	HF (36 model), MF (ovoz+SFX)

5. Qaysi tizimlar menda kuchli (saqlang!)

Bularni qayta yozmang — bu sizning raqobat ustunligingiz. Ikkala raqibda ham yo'q yoki zaifroq.

Tizim	Sizdagi ustunlik	Raqiblar
To'liq backend + DB	Express + Prisma + PostgreSQL + 7 migratsiya, tipli sxema	MF: WordPress · HF: yopiq SaaS
Auth xavfsizligi	PluginToken 32-bayt random, 30d, DB-saqlash, server-validatsiya	MF: date-token ddmmyy
Plan gating	Server-side checkDownloadAllowed — chetlab bo'lmaydi	MF: localStorage
Contributor workflow	DRAFT→upload→mogrt-extract→PENDING→review→APPROVED	Ikkalasida ham yo'q
Moderatsiya + audit	approve/hard-reject/soft-reject + StudioAuditLog + Resend email + thread	Ikkalasida ham yo'q
SSE realtime	Upload-progress stream (receive→sync→db→extract→done)	Ikkalasida ham yo'q
Storage yetukligi	R2 stream-upload, OOM-fix, 3GB fayl ~32MB peak	MF: CDN to'g'ridan-to'g'ri
Redirect xavfsizligi	Cross-origin'da Auth-header strip (token leak yo'q)	MF: yo'q
Ikki panel	Browse + Admin alohida	Ikkalasida ham bitta
TypeScript monorepo	npm workspaces, tipli, shared paketlar	Ikkalasi ham vanilla JS

6. Nima qo'shish kerak va NEGA — modullar

Ustuvorlik tartibida. Har biri: **nega** (texnik sabab) + **qaysi tech** + **qanday**.

1 ZXP build + imzolash quvuri

BLOCKER

Nega:

Imzolangan ZXP'siz oxirgi foydalanuvchi plaginni o'rnatish imkoniyati yo'q. Hozir faqat `install-cep.sh` (dev-only). Bu boshqa hamma narsadan oldin.

Qaysi tech:

Bolt CEP (`vite-cep-plugin` ZXP o'z ichida) YOKI `ZXPSignCmd` + self-signed `.p12`.

Qanday:

`scripts/build-zxp.sh` → sertifikat yaratish → `ZXPSignCmd -sign ./dist app.zxp cert.p12 pass` → tarqatish (yoki ZXP-installer).

2 Typed bridge (evalJSX) + @Sentry + @Refresh token

Nega:

evalJSX — injection xavfini tugatadi va type-safety beradi. Sentry — production ko'rinishi (hozir 0). Refresh — 30 kun tugaganda majburiy login'ni yo'qotadi.

Qaysi tech:

Bolt CEP evalTS (yoki qo'lda wrapper) · `@sentry/browser` + `@sentry/node` · JWT refresh endpoint.

Qanday:

`evalJSX(fn, ...args)` · JSON-only · Sentry init + sensitive-filter regex · `POST /api/plugin/refresh`.

3 Qidiruv mexanizmi

Nega:

`findMany` + JS-filter scale bo'lmaydi, typo-tolerance/relevance yo'q. Katalog o'sganda sekinlashadi.

Qaysi tech (bosqichma):

① PostgreSQL `tsvector` / `pg_trgm` (sizda Postgres bor — tekin) → ② Meilisearch/Typesense (self-host) → ③ Algolia (bepul 10k/oy).

Qanday:

ContributorTemplate'ga `searchVector` ustun + GIN index + `websearch_to_tsquery`. Faqat APPROVED+published indekslanadi.

4 AI Tools bo'limi (yangi modul + daromad)

Nega:

Mahsulot pozitsiyasi ("creative-tools"), raqobat pariteti, yangi daromad (kredit). Mavjud auth+subscription ustiga proxy sifatida o'rnatiladi.

Qaysi tech:

ElevenLabs (ovoz/SFX) · fal.ai/Replicate (rasm) · OpenAI embeddings (semantik qidiruv) — hammasi server-side proxy

Qanday:

`apps/api/src/routes/ai.ts` → auth → kredit-gate → provider-proxy (kalit serverda) → R2 → signed URL. Prisma: `aiGeneration` + `aiCredits`.
MF kabi kalitni klientga qo'ymang, exec(curl) ishlatmang.

5 Auto-update · @Feature flags · @Product analytics

Nega:

Update — tuzatishni yetkazish. Flags — AI'ni xavfsiz rollout. Analytics — data-asosli qaror.

Qaysi tech:

`GET /api/plugin/version` + semver · GrowthBook/PostHog flags · Amplitude/PostHog analytics.

8 Infratuzilma tuzatishlari

• **Stripe-bypass:**

`PLUGIN_ALLOW_PRO_WITHOUT_STRIPE=false` (hozir PRO bepul!).

• **Distributed state:**

ko'p-instansiyaga o'tranda Redis (rate-limit + SSE + cache) — hozir in-memory.

• **Dual asset model:**

`Asset/Download` (legacy) vs `ContributorTemplate` birlashtirish — maintenance qarzi.

- **Preview URL:**

public R2 — signed qilish yoki ataylab ochiq qabul qilish.

7. To'liq strategiya — maqsad arxitektura

Hozirgi kuchli poydevor + raqiblardan olingan eng yaxshi tizimlar = maqsad holat.



Qaror matritsasi — qaysi tizim qayerdan

Maqsad tizim	Manba	Harakat
Build + ZXP	HF (Bolt CEP)	Ko'chish — 3 muammoni hal qiladi
Typed bridge	HF	evalTS qabul qilish
Monitoring	HF	Sentry + filter regex
Feature flags	HF	GrowthBook/PostHog
Cache/polling	HF	Query-cache abstraktsiya
Qidiruv	MF	pg_trgm → keyin Algolia
Auto-update	MF	Versiya endpoint + semver
AI ovozi/SFX	MF formula	Server-proxy (xavfsiz)
Multi-app detection	MF	getHostEnvironment switch
Backend/DB/auth	SIZNIKI	Saqlash + sayqal
Contributor/moderatsiya	SIZNIKI	Saqlash (unikal ustunlik)
SSE/storage/gating	SIZNIKI	Saqlash + Redis (scale)

Yo'l xaritasi (texnik bosqichlar)

Bosqich	Tizimlar	Natija
0 (kritik)	ZXP build+sign · Stripe-bypass tuzatish	O'rnatish + daromad
1 (1-3 hafta)	Bolt CEP migratsiya → evalTS + ZXP + UI birga · Sentry	Poydevor
2 (3-5 hafta)	Refresh token · pg_trgm qidiruv · auto-update endpoint	Yetuklik
3 (1-2 oy)	AI Tools (semantik qidiruv → ovozi/SFX) · feature flags · analytics	Kengayish
4 (scale)	Redis · Algolia · multi-app (PPRO) · dual-model birlashtirish	Korxona

Yakuniy strategik xulosa

AssetFlow noyob holatda:

biznes-mantiq va backend raqiblardan kuchli

, lekin

plugin qatlami yetuk emas

. Strategiya — raqiblardan

biznes-modelini emas, balki texnik yetuklikni

olish: Higgsfield'dan framework+ko'prik+monitoring, Motion Factory'dan qidiruv+update+AI-formula, va o'zingizning kuchli poydevoringizni saqlash.

Eng yuqori richag — Bolt CEP migratsiyasi

, chunki u ZXP + typed bridge + zamonaviy UI'ni bitta harakatda beradi. Hech narsani qayta yozish kerak emas — faqat ustiga qurish.